

# PHISHING DETECTION REPORT

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## ANALYSIS OF EMAIL SAMPLES

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# 6 Emails

Total Emails Analyzed

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## Key Findings



2 Safe emails identified



2 Suspicious emails identified



2 Phishing emails detected

# EMAIL CLASSIFICATION SUMMARY

ALL ANALYZED SAMPLES AT A GLANCE

| # | Email Subject                                    | Sender                                                                           | Classification |
|---|--------------------------------------------------|----------------------------------------------------------------------------------|----------------|
| 1 | Security alert                                   | Google                                                                           | Safe           |
| 2 | Find out more about our updated Terms of Service | Google                                                                           | Safe           |
| 3 | Something EVERY Business Needs                   | <a href="mailto:2b1lf@msn.com">2b1lf@msn.com</a>                                 | Suspicious     |
| 4 | Get Paid For What You Know                       | <a href="mailto:63rlfplk4@aaaticketsource.com">63rlfplk4@aaaticketsource.com</a> | Suspicious     |
| 5 | CLIENTE PRIME - BRADESCO LIVELO...               | banco.bradesco@atendimento.com.br                                                | Phishing       |
| 6 | Verify Your Wallet Now...                        | <a href="mailto:support@pcpilrj.fzensk.com">support@pcpilrj.fzensk.com</a>       | Phishing       |

# Methodology Overview

## Tools and Techniques Used in Analysis

The methodology used for this report involved a comprehensive approach to analyzing phishing emails. We employed tools such as **Google Header Analyzer** and **MxToolbox** to analyze email headers for authentication protocols like SPF, DKIM, and DMARC. Each email was scrutinized not only for technical indicators but also through a manual review of sender domains and content to identify potential social engineering tactics. This dual strategy helps ensure a thorough understanding of the email's legitimacy.

In addition to the technical analysis, the report emphasizes careful content review. This involves looking for red flags such as **mismatched sender addresses**, suspicious links, and manipulative language designed to create urgency. Each email was categorized based on its risk level: Safe, Suspicious, or Phishing. This comprehensive methodology is essential for effective phishing detection and awareness, allowing organizations to better protect themselves from evolving threats in the digital landscape.

# Safe Emails

## Analysis of Legitimate Samples

The analysis of safe emails reveals crucial insights into legitimate communications. In our investigation, we identified two emails classified as Safe, both sent directly by Google. The first, a 'Security alert' notifying of a new account sign-in, showed full SPF, DKIM, and DMARC authentication passing, with a clean delivery path through Google's own mail servers. The second, a Terms of Service update notice, also passed all authentication checks, with every link routed through Google's official c.gle URL shortener. Both emails lacked any red flags typically associated with phishing, such as urgency, credential requests, or suspicious links. Both emails lacked any red flags typically associated with phishing, such as urgency, credential requests, or suspicious links, thereby reinforcing the importance of thorough examination in distinguishing safe emails from potential threats.

These safe emails serve as a benchmark for identifying legitimate communications. The 'Security alert' email, for instance, used calm, reassuring language ('If this was you, you don't need to do anything') rather than urgency. The Terms of Service update was purely informational and included a clear path to close the account if the user disagreed with the new terms. Both demonstrate what a legitimate, low-risk communication looks like: clean authentication, matching domains, and no manipulative language. Such communications reflect a **healthy digital landscape** where users can engage without fear of phishing attempts. Understanding the characteristics of safe emails is vital in developing a keen eye for identifying threats in one's inbox and fostering a more secure online environment.

# Suspicious Emails

## Analyzing Potentially Harmful Samples

In the realm of email security, **suspicious or spam emails** often employ tactics that can mislead recipients. These emails usually feature generic sender addresses and may include mass-blast patterns, indicating they are sent to multiple users. Common characteristics include vague promises, dubious links, and manipulative unsubscribe instructions. The aim is to elicit an action, whether that be clicking a link or responding with personal information, which can lead to further exploitation. Recognizing these patterns is crucial for maintaining security and awareness.

For instance, emails that request urgent actions, such as confirming account details or verifying information, often rely on **psychological manipulation**. They may offer enticing rewards or present dire consequences to prompt immediate reactions. Spam emails may also disguise themselves as legitimate communications through familiar branding or language. Analyzing sender domains and scrutinizing content for red flags, such as mismatched addresses or poor grammar, can significantly reduce the risk of falling victim to these types of threats. Awareness and vigilance are essential.

# Phishing Emails

## Analysis of Malicious Samples

Phishing emails are crafted to deceive recipients into revealing sensitive information. They often impersonate legitimate organizations, utilizing urgency and fear to prompt immediate action. In this report, we analyzed two distinct phishing emails that exemplify common tactics used by cybercriminals. Understanding these techniques is crucial for effective detection and awareness. By examining the content and the methods employed in these malicious samples, we aim to enhance awareness and improve defenses against phishing threats.

The first phishing email impersonated a well-known Brazilian bank, using urgency about expiring loyalty points to manipulate the recipient. The second email claimed to be a wallet verification request but masked its true intent through deceptive practices. Both emails failed multiple authentication checks, highlighting that legitimate-looking communications can still pose significant risks. These cases demonstrate the need for vigilance and analytical skills in recognizing and mitigating phishing threats effectively.

# Phishing Indicators

## Key Signs of Phishing Attempts

Common phishing indicators are critical for identifying suspicious emails. Look for **mismatched sender addresses**, which may not align with legitimate domains. Other indicators include mass-blast patterns, like **Undisclosed Recipients**, and urgency in language that creates fear. Emails that employ manipulative unsubscribe mechanisms or claim urgent action required, like account verification, are often phishing attempts. Recognizing these signs empowers users to remain vigilant against potential threats.

Additional details on common indicators include: failing authentication checks such as SPF, DKIM, or DMARC, which are essential for verifying email legitimacy. Be cautious of vague service identities and overly formal language that can indicate deception. Links that redirect through suspicious domains or use tracking services should raise red flags. Lastly, be wary of unsolicited attachments that could contain malware. Understanding these details can significantly enhance your email security awareness.



# Prevention Guidelines

## Essential Practices for Email Safety

The **Prevention and Awareness Guidelines** are essential for recognizing and mitigating phishing threats. Always verify the sender's actual email domain instead of relying on the display name. Hovering over links before clicking allows you to see the real destination. It's crucial to check header authentication results such as SPF, DKIM, and DMARC using tools like Google Header Analyzer or MxToolbox when in doubt. If an email claims urgent action, contact the organization directly through a known official channel to confirm its authenticity.

Additionally, reporting suspicious emails to your IT or security team enhances your organization's overall security posture. This proactive approach helps in identifying patterns of phishing attempts and can lead to more effective measures being put in place. Educating employees about these do's not only empowers them to be vigilant but also creates a culture of security awareness. Regular training sessions can reinforce these guidelines, ensuring that everyone understands the critical nature of email security and how to navigate potential threats.

# Prevention Don'ts

## Key Guidelines for Awareness

To effectively combat phishing threats, it's crucial to **avoid certain common pitfalls**. Don't trust an email solely because it has passed SPF, DKIM, or DMARC checks; these can be manipulated by anyone using free platforms. Moreover, never click on links or open attachments from unexpected emails, even if they appear to come from familiar senders. Additionally, avoid replying to spam to unsubscribe unless you're certain the source is legitimate, as this can confirm your address is active and lead to more unsolicited emails.

Always stay vigilant against urgency tactics; genuine organizations rarely demand immediate action via email. Don't assume a professional-looking email is automatically legitimate, as attackers often use polished designs to deceive their targets. If you feel something is off, trust your instincts. Implementing these simple yet effective prevention strategies can significantly reduce your risk of falling victim to phishing scams and enhance your overall email security posture.

# Conclusion

## Synthesizing Detection and Awareness

This exercise demonstrates that **phishing detection** requires both technical header analysis and content-based judgment. Authentication checks are a necessary but not sufficient signal of legitimacy. A combination of domain verification, content review, and **healthy skepticism** toward urgency and unnamed services provides the strongest defense against phishing attempts.

Understanding the intricacies of phishing attacks is essential for both individuals and organizations. By recognizing common tactics and indicators, users can become proactive in their defenses. Continuous training and awareness campaigns will empower users to spot potential threats and report them promptly, thus creating a safer digital environment for all.